

## 1. Architectural Choice

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System. The system is organized into Presentation, Business, and Data layers. The Presentation Layer handles touchscreen interactions, the Business Layer manages orders, payments and receipt printing, and the Data Layer stores menu and pricing information.

## 2. Reason 1 – Separation of Concerns

Layered Architecture provides a clear separation of responsibilities between different parts of the system. The User Interface handles customer interaction, the Order Manager handles order processing, the Payment Service handles credit-card payments, the Receipt Printer handles receipt generation, and the Database manages menu and pricing information. This makes the system easier to understand and maintain.

## 3. Reason 2 – Simplicity

The Self-Service Coffee Kiosk has a relatively straightforward workflow involving coffee selection, size selection, payment and receipt printing. Layered Architecture is therefore appropriate because it provides the required organization without introducing the additional operational complexity associated with Microservices Architecture.

## 4. Security Advantage

Layered Architecture allows payment-related functionality to be separated into the dedicated **Payment Service Component**. This prevents the User Interface from directly handling all payment-processing responsibilities and provides a clear separation between customer interaction and payment processing.

## 5. Performance Benefit

Separating the system into layers allows each component to perform a specific responsibility. The Order Manager can coordinate order processing while the Database provides menu and pricing information when required. This organized communication helps avoid unnecessary processing in the User Interface and keeps system interactions efficient.

## Selected Architecture: Layered Architecture

**Components:** User Interface, Order Manager, Payment Service, Receipt Printer, Database